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Unit 2 Class Activity 1: 20.07.2026

1. Hospital Patient Admission Analysis

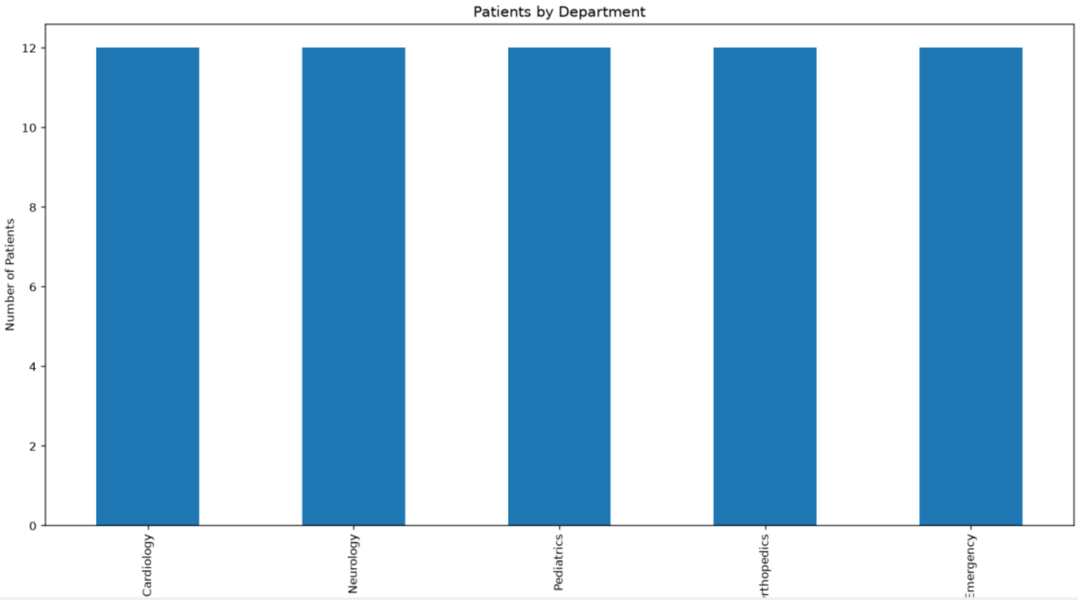
A hospital has collected data on patient admissions across multiple departments (Cardiology, Neurology, Pediatrics, Orthopedics, Emergency) for the past year. Different visualizations (bar chart, stacked bar chart, treemap, box plot, histogram) are available. Evaluate which visualization most effectively communicates department workload, admission variability, and patient distribution. Justify your selection with evidence.

Python code:

```
import pandas as pd

import matplotlib.pyplot as plt


# Read the CSV file

df = pd.read_csv("hospital_patient_admission.csv")


# Display first 5 records

print(df.head())


# Bar Chart - Patients by Department

df["Department"].value_counts().plot(kind="bar")

plt.title("Patients by Department")
```

```
plt.xlabel("Department")

plt.ylabel("Number of Patients")

plt.show()


# Histogram - Length of Stay

plt.hist(df["Length_of_Stay"], bins=5, edgecolor="black")

plt.title("Length of Stay")

plt.xlabel("Days")

plt.ylabel("Patients")

plt.show()


# Box Plot - Age

plt.boxplot(df["Age"])

plt.title("Patient Age")

plt.ylabel("Age")

plt.show()


# Analysis

print("\nAverage Age:", df["Age"].mean())

print("Average Length of Stay:", df["Length_of_Stay"].mean())
```

Output:



Patients by Department
Number of Patients
12
10
8
6
4
2
0
Cardiology
Neurology
Pediatrics
rthopedics
Emergency

2. E-Commerce Customer Spending Behaviour

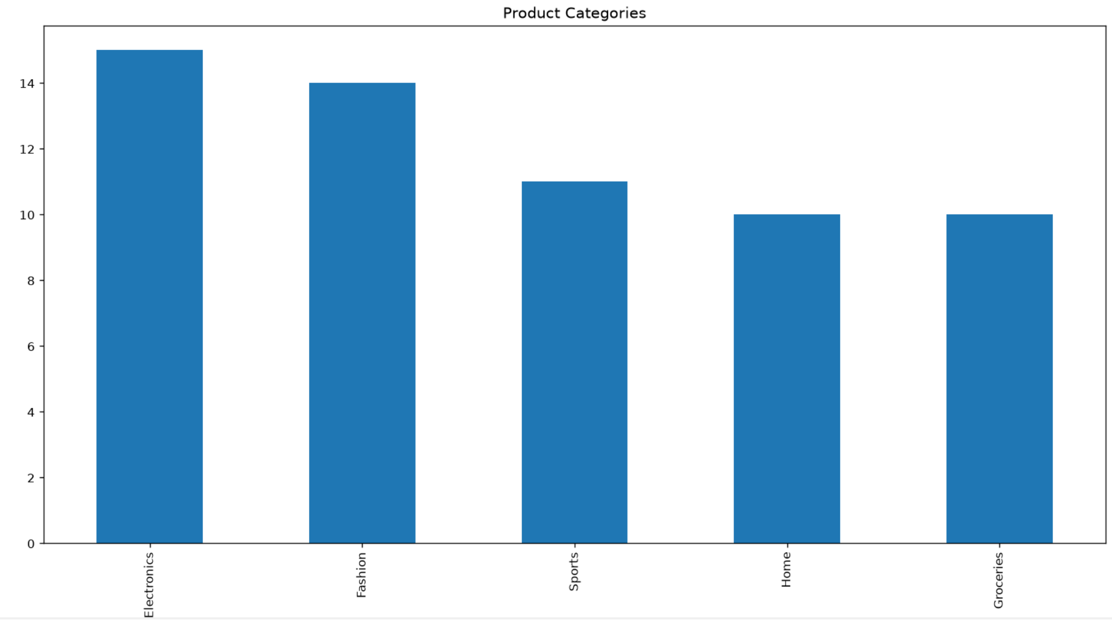
An online retailer has customer purchase amounts, order frequency, and product categories. Create multiple visualizations (histogram, violin plot, box plot, density plot, bar chart). Evaluate which visualization best identifies high-value customers, spending variability, and purchasing trends. Recommend the most appropriate visualization for business executives and explain why.

Python code:

```
import pandas as pd

import matplotlib.pyplot as plt



df = pd.read_csv("ecommerce_customer_spending.csv")



print(df.head())



df["Category"].value_counts().plot(kind="bar")

plt.title("Product Categories")

plt.show()



plt.hist(df["Amount_Spent"], bins=5, edgecolor="black")

plt.title("Amount Spent")

plt.show()
```

```
plt.boxplot(df["Amount_Spent"])

plt.title("Amount Spent")

plt.show()
```

Output:



3. University Examination Performance Dashboard

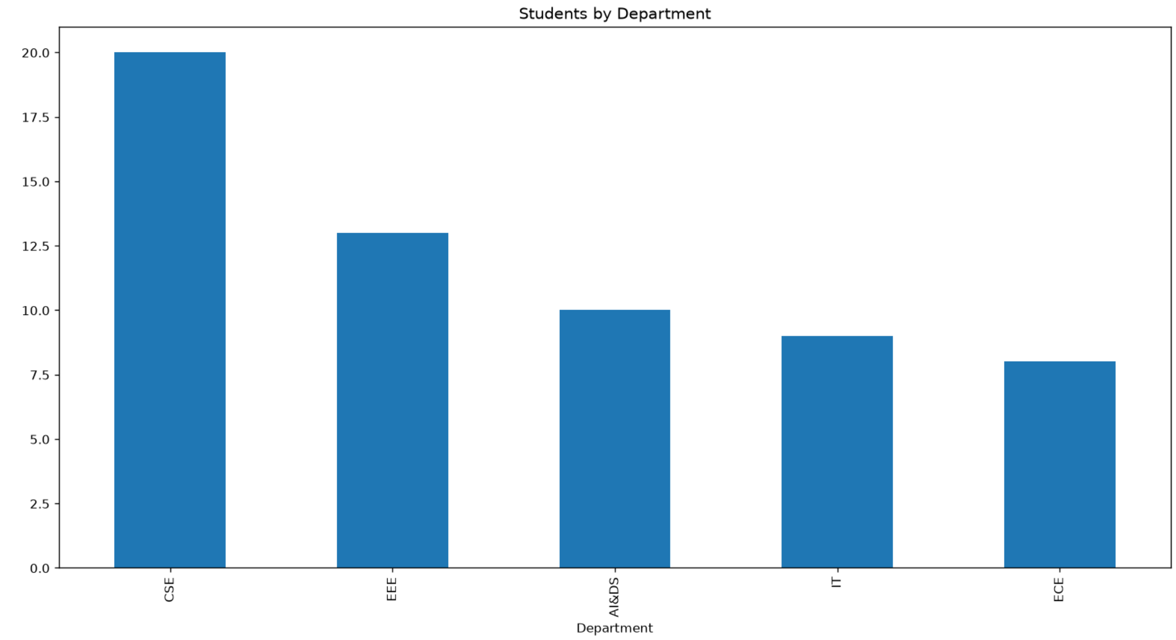
A university wants to compare student marks across departments and identify score distributions, outliers, and top-performing classes. Generate different visualizations and evaluate which charts provide the clearest insights for academic decision-making. Recommend improvements to the visualization dashboard.

Python code:

```
import pandas as pd

import matplotlib.pyplot as plt



df = pd.read_csv("university_exam_performance.csv")



print(df.head())



df["Department"].value_counts().plot(kind="bar")

plt.title("Students by Department")

plt.show()



plt.hist(df["Final_Marks"], bins=5, edgecolor="black")

plt.title("Final Marks")

plt.show()
```

```
plt.boxplot(df["Final_Marks"])

plt.title("Final Marks")

plt.show()
```

Output:



4. Climate and Rainfall Distribution Study

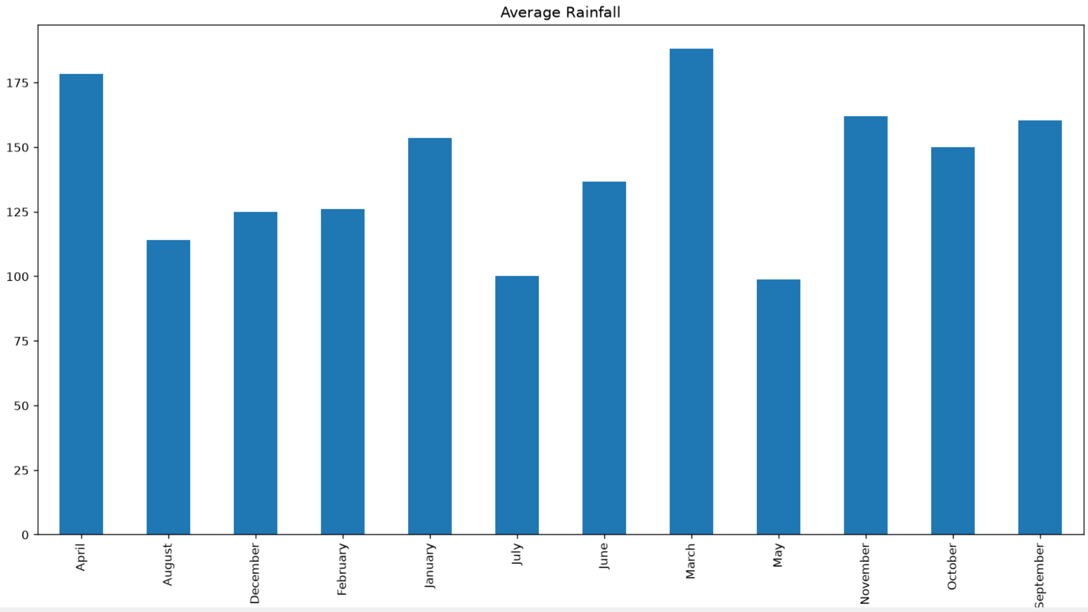
Meteorological data contains monthly rainfall, temperature, and humidity for multiple cities. Produce histograms, density plots, box plots, and bar charts. Evaluate which visualization best represents seasonal rainfall variation and climate distribution. Compare at least three visualizations and justify the most suitable one for environmental researchers.

Python code:

```
import pandas as pd

import matplotlib.pyplot as plt



df = pd.read_csv("climate_rainfall_distribution.csv")



print(df.head())



df.groupby("Month")["Rainfall_mm"].mean().plot(kind="bar")

plt.title("Average Rainfall")

plt.show()



plt.hist(df["Rainfall_mm"], bins=5, edgecolor="black")

plt.title("Rainfall")

plt.show()
```

```
plt.boxplot(df["Rainfall_mm"])

plt.title("Rainfall")

plt.show()
```

Output:



5. Bank Loan Risk Assessment

A financial institution has customer loan amounts, repayment history, income levels, and credit scores. Develop visualizations showing loan distributions and customer segments. Evaluate which visualization best supports loan approval decisions by highlighting risk patterns, skewness, and outliers. Recommend the visualization that should be included in a management dashboard.

Python code:

```
import pandas as pd

import matplotlib.pyplot as plt



df = pd.read_csv("bank_loan_risk.csv")



print(df.head())



df["Employment_Type"].value_counts().plot(kind="bar")

plt.title("Employment Type")

plt.show()



plt.hist(df["Credit_Score"], bins=5, edgecolor="black")

plt.title("Credit Score")

plt.show()
```

```
plt.boxplot(df["Loan_Amount"])

plt.title("Loan Amount")

plt.show()
```

Output: